

Decentralized Privacy-Preserving Federated Medical AI with Blockchain and Selective LoRA

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Abstract—The introduction of AI in healthcare has led to some ethical issues like data security, patient privacy and reliability. In addition, centralized data is vulnerable to cyber-attacks in every field. Considering these issues model should be improved without sharing patient’s raw data. Federated learning approach can be used whereas the model trains on edge devices and passes its updated parameters. The central server collects and aggregates different client’s updated parameters to improve the model. The system uses selective LoRA concepts to reduce training cost. Introducing multiple servers so that the client can randomly select a server to send its updates. But now also the system is centralized. For that the architecture implements blockchain technology that helps the system to log the servers’ activities. The same method can also be implemented on the client side for client-to-client communication. A scoring methodology is initiated based on client’s participation, contribution and status of availability. Differential Private Retrieval-Augmented Generation (DP-RAG) is used in the client side for information retrieval and knowledge generation. Tensor tagging mechanism is implemented in the system so that the server gets an additional layer of defense while processing clients’ updates. Using the gRPC technology for communication the system becomes faster.

Keywords—*Federated Learning, Medical AI, Blockchain, Selective LoRA, Differential Privacy, Retrieval-Augmented Generation, Secure Model Update, gRPC*

I. INTRODUCTION

The introduction of artificial intelligence brought a significant change in the healthcare field. Patient management and diagnosis became much more efficient. But with this efficiency some problems like data security, patient privacy and reliability are raised. To understand this situation, we can consider a scenario where a model on the central server is updated only when raw data is transferred from users. But in medical field each data is valuable and crucial. So, patients do not encourage passing of raw data to an unknown AI model. This leads to the concept of federated learning. A model is trained with private data in the users’ edge device such as mobile, personal computer, etc. and only the updated parameters passed to the central server. With these updated weights from different devices, the central server aggregates them to form a global model. This method is called federated learning. Here the global model learns or is fine-tuned without seeing the original data. But this method has a drawback. The training process is costly, time consuming and requires high-end device.

To eliminate this issue another methodology called LoRA or Low-Rank adaptation is used. With this method the system can fine-tune only the selective parameters of a model rather than updating the whole. This technique helps enable training on any device regardless of their hardware availability. The client with more efficient hardware can fine-tune the model with large number of selected layers whereas smaller device can also participate by reducing the number of selective parameters. The server aggregates the parameters and makes a global LoRA. The global LoRA can also attach to the client's model.

The system is centralized and it has its own demerits. In this client-server architecture, the client has only one option to send the updated lora weights. So, by providing multiple servers at the same time the client can randomly select which server to use. The server activities are recorded by a blockchain mechanism like the communication of servers, the updates that they are doing and many more. This helps to continuously log the servers' details. As this is a centralized system, client-to-client communication is not possible. If the system allows this feature, an attacker can target any client, and evidence of the attack may be lost. So, the same blockchain methodology can be used and when a client wants to participate in client-to-client communication, they should register to the blockchain. If all the servers get attacked, the system does not shut down completely. The client-to-client communication blockchain still exists and the server can take necessary things like the global LoRA.

The system cannot trust all the clients, as there may be attackers among them. These clients train their local model with miscellaneous data and try to perform poisoning attacks. So, the system classifies all incoming tensors as safe, sensitive and risky. But we can't reject the sensitive and risky tensors because medical field is evolving and there might be some special cases and rare diseases. Because of that the sensitive and risky tensors undergo for some kind of validation and decide whether it is an anomaly or an attack. This type of system is scalable only when there are more clients. Clients do not participate in the architecture when there is no benefit. First, a heartbeat mechanism is set. The client periodically sends a signal to the server including that they are alive. Based on the heartbeat, client contribution and some other factors, a score is given to the client. This score will increase and decrease based on client activities.

A hybrid encryption is done with symmetric and asymmetric encryption methodologies. In the system, the updated parameters are encrypted with a symmetric encryption method. Then the key of symmetric encryption is encrypted using the server's public key. This is because the computational cost of asymmetric encryption is high. The key and the encrypted message are then sent to the server through different channels. For communication the architecture is using gRPC, Google Remote Procedure Call. The gRPC uses HTTP 2 and protocol buffers, which is more reliable and faster than normal API communication. Some patients have rare diseases, and their data is required for doctors and research centres. For that, an internationally accepted format, FHIR, is used for data portability. Differentially private Retrieval-Augmented Generation helps retrieve information of clients from a locally stored vector database and generate appropriate results for the user query. This measure uses controlled noises while adding data to the vector database so that the reconstruction of knowledge without decoding may lead to error.

II. RELATED WORK

Federated learning is an effective way to train machine learning models without sharing raw data. The work by McMahan et al. [1] introduced a federated averaging method (FedAvg), it allows model training by aggregating updates from different clients. This method focuses on the data size aggregations but does not consider the reliability and trustworthiness of clients. A comprehensive survey of open challenges in federated learning, including privacy, robustness and communication efficiency, is presented in [6].

To improve the communication efficiency, E. J. Hu et al. [2] introduced Low-Rank Adaptation of Large Language Models. In this method only small number of parameters are trained instead of the whole model. The LoRA made training possible even in resource limited devices. While LoRA reduces computational cost, existing works do not address how these updates can verify in a distributed and secured manner.

Retrieval-Augmented Generation (RAG) introduced by Lewis et al. [3] improved response quality in information retrieval. This reduces hallucination of generative models. But in medical data, RAG can expose sensitive patient data if we don't take proper privacy measures. Differential privacy, which is introduced by Dwork [4] and extended to deep learning by Abadi et al. [5], proposed a mathematical framework to protect sensitive data. It may affect model utility if not properly managed. Existing methods

treat privacy and model performance as separate. Some recent works integrated federated learning in healthcare [11], [12]. In these studies, collaboration across institutions is required. This approach tells the importance of privacy preserving in healthcare but do not address the issue of unreliable clients.

Blockchain-based mechanisms were also proposed in [8], [9], [10] to improve security and transparency in distributed system. A comprehensive survey of blockchain-based federated learning is presented in [22]. Blockchain has also been applied specifically to secure federated learning frameworks [23]. These approaches enable decentralized transparency. In a distributed server network, clients can randomly select a server to transmit updates. But they are not tightly integrated with model validation and scoring mechanism. Existing approaches address individual aspects such as privacy, efficiency, reliability and decentralization. But there is no unified framework for achieving all of this. The proposed system addresses this gap in healthcare field and tries to integrate selective LoRA, decentralized distributed system using blockchain, effective and secure communication, scoring mechanism considering reliability and constancy, differential privacy and portability of structured data.

Table I: advantages and disadvantages of already existing methods

Method Proposed	Advantages	Disadvantages
FedAvg: Aggregates model updates from distributed clients weighted by local data size [1]	Enables collaborative training without sharing raw data; scalable across large numbers of clients; supports heterogeneous data distributions	Aggregation based only on data size; does not consider client reliability or trustworthiness; vulnerable to poisoned updates from malicious clients
LoRA: Fine-tunes only injected low-rank matrices instead of full model weights for efficient adaptation [2]	Drastically reduces trainable parameters; enables fine-tuning on resource-constrained edge devices; lowers communication overhead	No mechanism to verify or validate low-rank updates in a distributed or secure manner; susceptible to adversarial manipulation of rank decomposition
RAG: Augments LLM responses by retrieving relevant documents from an external index at inference time [3]	Reduces hallucination; improves factual accuracy without full retraining; supports dynamic knowledge updates with minimal cost	Risk of exposing sensitive patient records in retrieved context; lacks privacy-preserving retrieval mechanisms for medical data; retrieval quality depends on index freshness
Differential Privacy: Mathematical framework adding calibrated noise to query outputs to protect individual-level data [4, 5]	Provides provable, mathematically rigorous privacy guarantees; applicable to query-based systems and statistical learning; composable across mechanisms	Noise addition degrades model utility, especially on small datasets; privacy-utility trade-off is hard to tune; does not protect against all inference attacks
Federated Learning in Healthcare: Multi-institutional collaboration enabling joint model	Enables cross-institutional learning without centralizing sensitive patient data; improves model	Does not address unreliable or malicious clients; lacks scoring or incentive mechanisms;

Method Proposed	Advantages	Disadvantages
training across hospitals [11, 12]	generalization across diverse demographics	communication bottlenecks in large hospital networks
Blockchain for Security & Transparency: Decentralized ledger used to log and audit data access in distributed systems [8, 9, 10]	Provides immutable audit trails; enhances trust in multi-party systems; reduces single point of failure; supports tamper-evident logging	High computational and storage overhead; not tightly integrated with model validation or scoring; latency issues for real-time federated updates
Byzantine-Robust Aggregation: Defends federated learning against adversarial or faulty client updates using statistical filtering [17, 18, 19]	Improves resilience to model poisoning attacks; filters outliers from aggregation; enables safer deployment in adversarial environments	Computational overhead increases with number of clients; may incorrectly penalize legitimate clients with heterogeneous data; not designed for healthcare-specific data distributions
Secure Aggregation: Cryptographic protocol ensuring server cannot inspect individual client updates during aggregation [7]	Protects update privacy from the aggregation server; prevents inference of local training data; compatible with standard federated learning pipelines	Significant communication and computation overhead; requires synchronized participation; fails if too many clients drop out mid-round
Reputation-Based Federated Learning: Assigns trust scores to clients based on historical contribution quality to guide aggregation [24, 25]	Reduces influence of unreliable or malicious participants; improves model quality over time through adaptive weighting; enables incentive-compatible participation	Cold-start problem for new clients; reputation scores can be gamed; lack of standardized metrics for trustworthiness across domains

III. PROPOSED METHODOLOGY

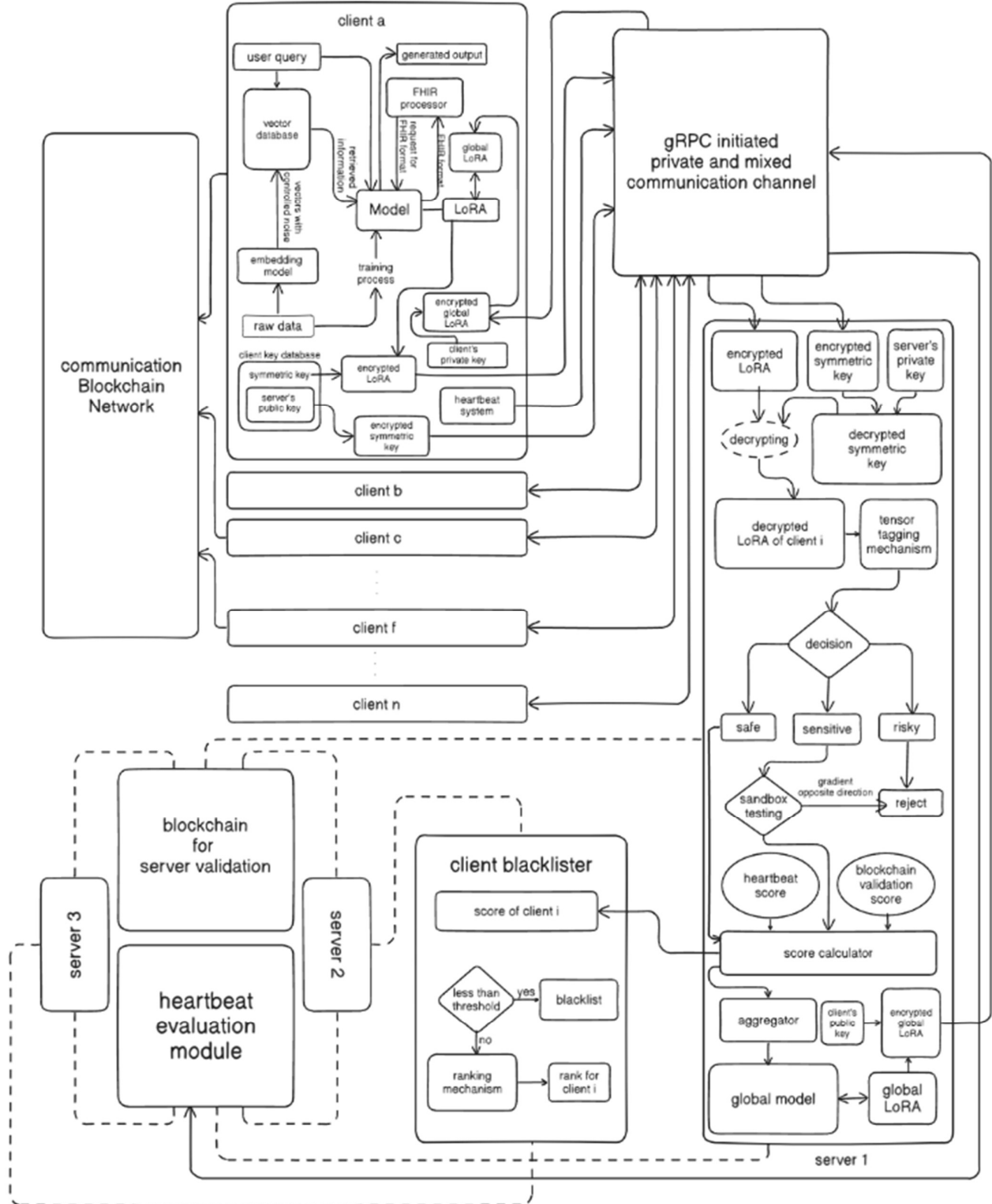


Fig. 1: System Architecture Illustrating Decentralized Federated Learning with Blockchain Integration, Selective LoRA Adaptation, and Secure Communication

The system, privacy preserving federated learning with blockchain, proposes a hybrid design that combines efficient model adaptation, secure communication and reliable validation. The system operates

in distributed clients and servers connected with blockchain for maintaining client data privacy, trust and robustness in learning process.

A. Selective LoRA-Based Model Adaptation

The system uses Low-Rank Adaptation to train only the selected parameters in the model instead of updating whole. This enables training and fine-tuning of model in any device irrespective of their hardware.

$$\begin{aligned}
& \text{Layers, } L = l_1, l_2, l_3 \dots, l_n; \text{ Weight for each layer, } W^{(l)}_{l \in L} \\
& \text{Low-Rank Adaptation for each layer, } W'^{(l)} = A^{(l)} \times B^{(l)}_{l \in L} \\
& \text{Updated weight per layer, } W^{(l)}_{new} = W^{(l)} + A^{(l)} \times B^{(l)}_{l \in L} \\
& \text{Full model output, } y_i = f(x; \{W^{(l)} + A^{(l)} \times B^{(l)}\}_{l \in L})
\end{aligned} \tag{1}$$

The formula (1) follows the principle of Low-Rank Adaptation introduced in [2]. Instead of updating whole parameters $W^{(l)}$, the updated matrix is factorized into two small matrices $A^{(l)}$ and $B^{(l)}$ belongs to $\mathbb{R}^{d \times r}$ and $\mathbb{R}^{r \times d}$, where $r \ll d$. This reduce the number of trainable parameters from $O(d^2)$ to $O(dr)$. The A and B are learning from patient data. Instead of transmitting full updates, the client sends only the LoRA metrics:

$$U_i = A^{(l)}, B^{(l)}_{l \in L}$$

Unlike standard LoRA [2], the proposed system introduces a new factor L_k , which adjust the number of layers based on the client's resource capacity.

$$\Delta W_k = L_k \cdot (A_k B_k) \tag{2}$$

B. Secure Communication using gRPC and Hybrid Encryption

The system ensures a hybrid encryption approach for conserving the confidentiality of transmitting data. The updated LoRA parameters are encrypted using a symmetric key and the key itself encrypted using the server's public key. This method helps to reduce the computational cost of cryptography. The encrypted data and the key can be transmitted through separate channels.

Conventional communication methods such as API can be less effective and slower. gRPC (Google Remote Procedure Call) is used for high performance. It uses HTTP/2 for low-latency data transmission and employs Protocol Buffers for data serialization.

The client can also request the global LoRA from any server. While attaching the global LoRA helps to generalize the output generated in the client side. This communication also uses the same mechanism. The hybrid encryption mechanism follows the design of secure aggregation protocol [7], which reduces encryption overhead from $O(n^2)$ to $O(n)$ in distributed system.

C. Heartbeat-Based Consistency Measurement

Each client periodically sends a signal to the server indicating participation. A minimum of H_{min} signal is expected every hour (If the periodic gap is 20 then at least 3 signals). The number of actual signals can indicate as H_i . So, the consistency score:

$$C_i^{(h)} = \min\left(1, \frac{H_i}{H_{min}}\right) \tag{3}$$

Frequency alone does not capture reliability. So, a stability component is added to evaluate the regularity. t_k is the set of actual time gaps between the heartbeat signals. t is the expected interval. The number of signals denoted by m .

$$S_i^{(t)} = 1 - \frac{1}{m} \sum_{k=1}^m \frac{|t_k - t|}{t}$$

$$S_i^{(t)} = 1, \text{ means perfect timing.} \tag{4}$$

The final consistency score is:

$$C_i = C_i^{(h)} \times S_i^{(t)} \quad (5)$$

This formula ensures both frequency and stability, thereby enabling client reliability. The stability term equation (4) is derived from [14 - 16] used in physiological signal analysis.

D. Tensor-Based Risk Evaluation

Each client update should be verified before being aggregated to the main model. Otherwise, the hackers can initiate poisoning attacks on the server. For this, a risk score is computed using magnitude and direction of the client's upload. The updated tensor is defined as $\Delta W_i = A^{(l)} \times B^{(l)}$

$$\text{magnitude}, M_i = \|\Delta W_i\| \quad (6)$$

The magnitude term M_i is used in gradient norm analysis in federated learning [18] which captures the strength of the updates. The D_i is derived from cosine similarity which is widely used in Byzantine-robust aggregation methods [17], [18].

$$\text{Direction}, D_i = \frac{\Delta W_i \cdot \Delta W}{\|\Delta W_i\| \|\Delta W\|} \quad (7)$$

where, $\Delta W = \text{global average}$

$$\text{Risk score}, R_i = \alpha M_i + \beta (1 - D_i) \quad (8)$$

Both magnitude and direction is essential for trust-based aggregation like FLTrust [19]. α and β are weights that can be given to magnitude and direction. The sum of these two must be 1. The value can be given based on the priority of magnitude and direction. From this we can calculate the quality of client update:

$$Q_i = 1 - R_i$$

Based on the R_i the tensors can be tagged as:

$$R_i \leq T_1 : \text{Safe}; T_1 < R_i \leq T_2 : \text{Sensitive}; R_i > T_2 : \text{Risky}$$

T_1 and T_2 are threshold values

Sandbox testing can be done before rejecting the sensitive and risky tensors follows principles of anomaly detection [20].

E. Blockchain-Based Validation by Servers

A set of servers which are connected by a blockchain network is used for validation. The client has the option to randomly select the server for sending its updates. Because all the servers are connected to a blockchain network, at least half of the servers should accept the client's update. Each server evaluates the client updates independently and votes for it.

$$V_i = \frac{n_i^{\text{valid}}}{N} \quad (9)$$

Where N is the number of servers participating from the blockchain network and n_i^{valid} is the number of servers that accept the update. The validation ratio in (9) is inspired by consensus mechanisms in distributed systems [21].

F. Client Scoring Mechanism

The final score for each client can be computed by combining consistency, quality and validation agreement. The scoring function is formulated as a weighted linear combination, like reputation-based systems in distributed environments [24], [25].

$$\begin{aligned} S_i^t &= \lambda_1 C_i + \lambda_2 (1 - R_i) + \lambda_3 V_i \\ S_i &= S_i^{(t-1)} + S_i^t \end{aligned} \quad (10)$$

λ_1 , λ_2 and λ_3 are weights for consistency, quality and validation respectively and its sum should be equal to 1. The score reflects the reliability of each client. $S_i^{(-1)}$ is the previous score.

G. Client Blacklisting

Clients are ranked based on their score, and a percentile-based mechanism is used to blacklist the unreliable participants. A threshold θ_p is set and clients have score below this probability threshold will be blacklisted.

$$\text{Blacklist if } P_i > \theta_p$$

This approach avoids fixed threshold mechanism for blacklisting.

H. Trust-Aware Aggregation Mechanism

Federated learning typically performs based on the number of data samples available for each client. This helps to give clients weight. However, this approach does not work for the reliability or quality of client updates. To overcome this limitation, the proposed system introduces a trust-aware mechanism. Each client k has a data size of n_k and score S_k , where the score is derived from consistency, quality and validation agreement of servers.

$$f(w) = \sum_{k=1}^K \frac{n_k S_k}{\sum_{j=1}^K n_j S_j} F_k(w) \quad (11)$$

In the proposed system instead of aggregate full model update, LoRA-based updates are used. For each client k :

$$\Delta W_k = A_k B_k \quad (12)$$

So, the global update becomes

$$\Delta W = \sum_{k=1}^K \frac{n_k S_k}{\sum_{j=1}^K n_j S_j} \Delta W_k \quad (13)$$

This approach provides a balanced aggregation strategy where both quality and quantity is measured. In real-world federated learning, same client can send multiple LoRA to the same server. This may cause dominance of the frequently sending clients. To resolve this a temporal time-based weighting mechanism is introduced.

$$w_i = e^{-\lambda(T - t_i)} \quad (14)$$

Where t_i is the timestamp of i th update, T is the current aggregation time and λ control the decay rate. Where n_k is the number of updates sent by a client k , then the final weight is

$$\Delta W_k^{final} = \frac{\sum_{i=1}^{m_k} w_i \cdot \Delta W_{i,k}}{\sum_{i=1}^{m_k} w_i} \quad (15)$$

Final global update

$$\Delta W = \sum_{k=1}^K \frac{n_k S_k}{\sum_{j=1}^K n_j S_j} L_k \Delta W_k^{final} \quad (16)$$

I. Differentially Private Retrieval-Augmented Generation (DP-RAG)

This functionality is aimed to be implemented on the client side that information storing and retrieving for client to analyze their medical condition. The information is stored in the vector database and query-based retrieval is initiated. The retrieval component is based on dense passage retrieval techniques [27]. This retrieved information and query is then treated to a generative model so that the model only generates output based on the available knowledge. This helps to eliminate the hallucination problem.

Before storing the data into the vector database, add a controlled amount of noise. Which helps to increase the difficulty of the reconstruction of data. More noise leads to the breaking of semantics of the information. When there is no relevant information in the database, the model uses global LoRA for generating a generalized result with a note saying that this output is not based on the client's record but may be relevant.

J. FHIR-Based Data Structuring and Interoperability

Sometimes patient data is required for doctor and research center if the patient's disease is rare. So an internationally accepted resource sharing format named FHIR (Fast Healthcare Interoperability Resources) [13], [28] is used. This system can convert patient's data into FHIR format and uses a secure private channel for transportation.

IV. EXPERIMENTAL SETUP

This section evaluates the client scoring mechanism and determine whether the score follows a known probability distribution. This experiment is conducted in a large-scale federated environment simulation.

A minimum number of 20 clients are taken with 3 distributed servers. Each client undergoes some rounds of LoRA training to ensure sufficient variation in client behaviour. The communication between client and server is intentionally made non-uniform so that it simulates a real-world scenario. For example, a server can get 2 or more updates of the same client in a single aggregation window, while another one get significantly fewer or nothing.

To evaluate the robustness of scoring, a subset of clients intentionally transmit malicious data. This process allows the system to ensure the working of tensor tagging mechanism.

After each training round, the client score is collected and analysed. After performing some empirical studies using these scores, we can evaluate whether the distribution of score comes under a known probability distribution. This helps to analyse the statistical behaviour of the scoring system. Based on the observed distribution a thresholding strategy can be derived to reduce the computational overhead.

Instead of recalculating the ranking of each one when there is a change in the number of active clients due to heartbeat signals, the system can use distribution informed threshold. This reduces the need for repeated computation. This experimental setup validates the effectiveness of scoring mechanism and find whether the score follows a known probability distribution which helps to reduce the computational cost.

After collecting all scores from 1000 events, statistical analysis was performed to identify a suitable percentile-based blacklisting threshold. Standard statistical methods for outlier detection [26] were used to inform the thresholding strategy. The Shapiro–Wilk normality test was applied to a representative sample of 200 events to assess the distributional behaviour of the scores.

V. RESULT

The experiment was conducted to evaluate the effectiveness of the proposed system in a real-world scenario. It helps to identify that the system is trust-aware and a realistic aggregation strategy. Through this experiment we found a threshold that can be used to blacklist the unreliable clients participating in the system. Global model present in the system will not get affected by the participation of the unreliable client in the aggregation phase.

A. Score Distribution Analysis

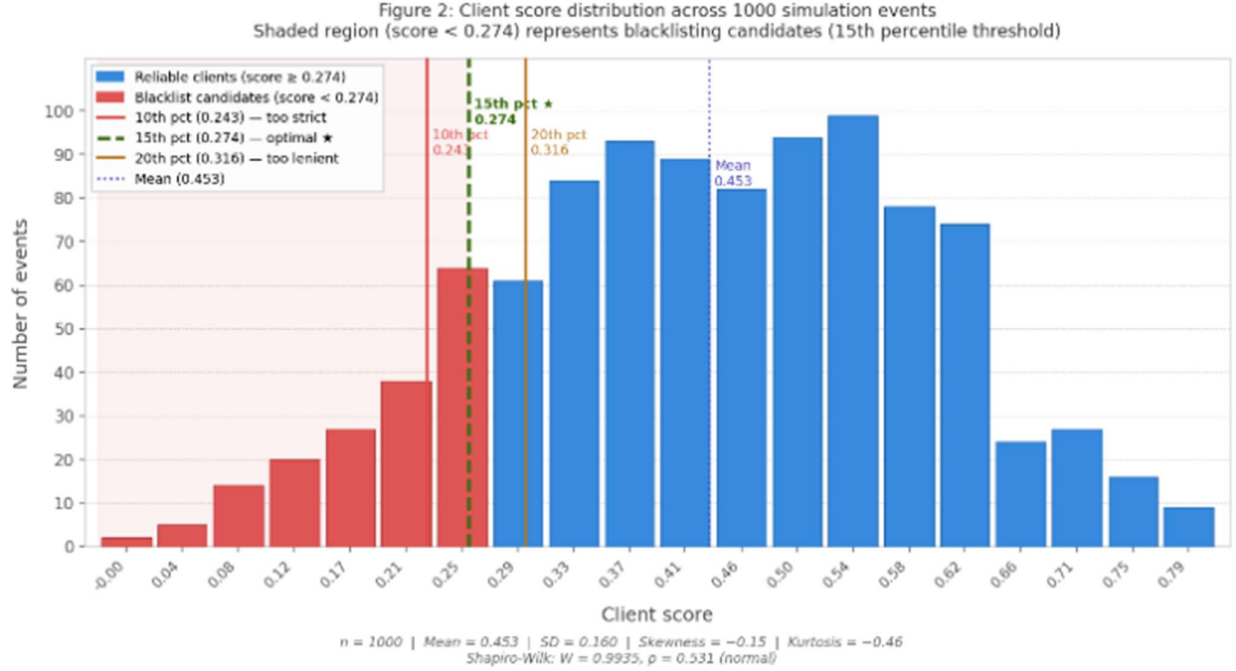


Fig. 2: client score distribution of 1000 events. The red shaded region has a score less than 0.274 identifies the blacklisting candidate identified at the 15th percentile threshold.

Through this graph we can see that the score ranges from -0.002 to 0.831. The mean and standard deviation are 0.453 and 0.160 respectively. The distribution exhibits a slightly left-skewed, platykurtic shape (skewness = -0.15, kurtosis = -0.46), indicating a broad spread of client behaviours with limited extreme outliers. The Shapiro–Wilk normality test confirmed that the distribution does not significantly deviate from normality ($W = 0.9935$, $p = 0.531$), supporting the use of distribution-based thresholding. Based on the percentile analysis, the 15th percentile is selected as the optimal blacklist boundary ($\theta = 0.274$). Only 106 events belongs to this threshold range.

Parameter	Value / Observation	Description / Notes
Total events	1000	Number of simulation events taken for analysis
Mean score	0.453	Average client score
Standard deviation	0.160	Variation in scores
Skewness	-0.15	Slightly left-skewed distribution
Kurtosis	-0.46	Flatter distribution compared to normal
Minimum score (approx.)	-0.002	Lower bound of score range
Maximum score (approx.)	0.831	Upper bound of score range
10th percentile	0.243	Strict threshold (too rigid)
15th percentile (selected)	0.274	Chosen blacklist threshold
20th percentile	0.316	Lenient threshold (too relaxed)
Reliable region	Score ≥ 0.274	Clients considered reliable
Blacklist region	Score < 0.274	Clients marked for potential blacklisting
Shape of distribution	Slightly left-skewed, platykurtic	Most scores concentrated around middle with fewer extreme high values
Blacklisting candidates	106 events	Events below 15th percentile threshold

Table 2: Summary of score distribution characteristics

B. Heartbeat Consistency Behaviour

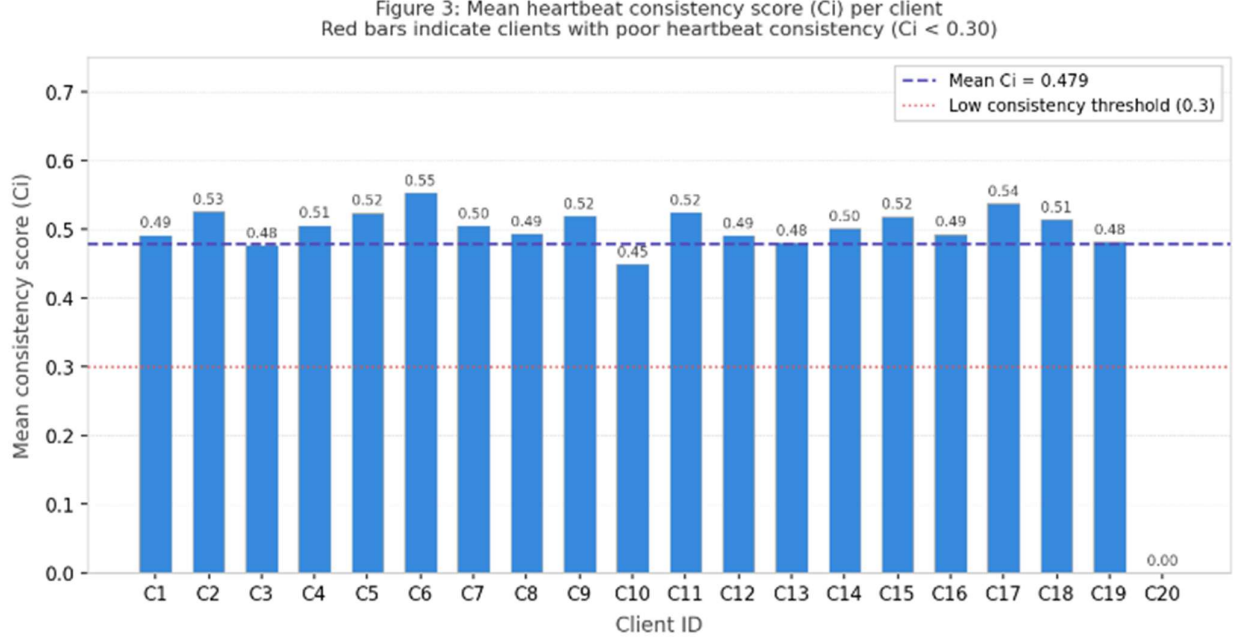


Fig. 3: mean heartbeat consistency score for each client. Red bar indicate clients with poor heartbeat regularity ($C_i < 0.30$)

Fig. 3 explains that the client C20 exhibit a consistency score of 0.0, which means the client is unreliable. From the simulated experiment the average consistency score was 0.479 across all clients. This confirms that Section III-C has a significant role for checking client reliability.

C. Tensor Risk Analysis

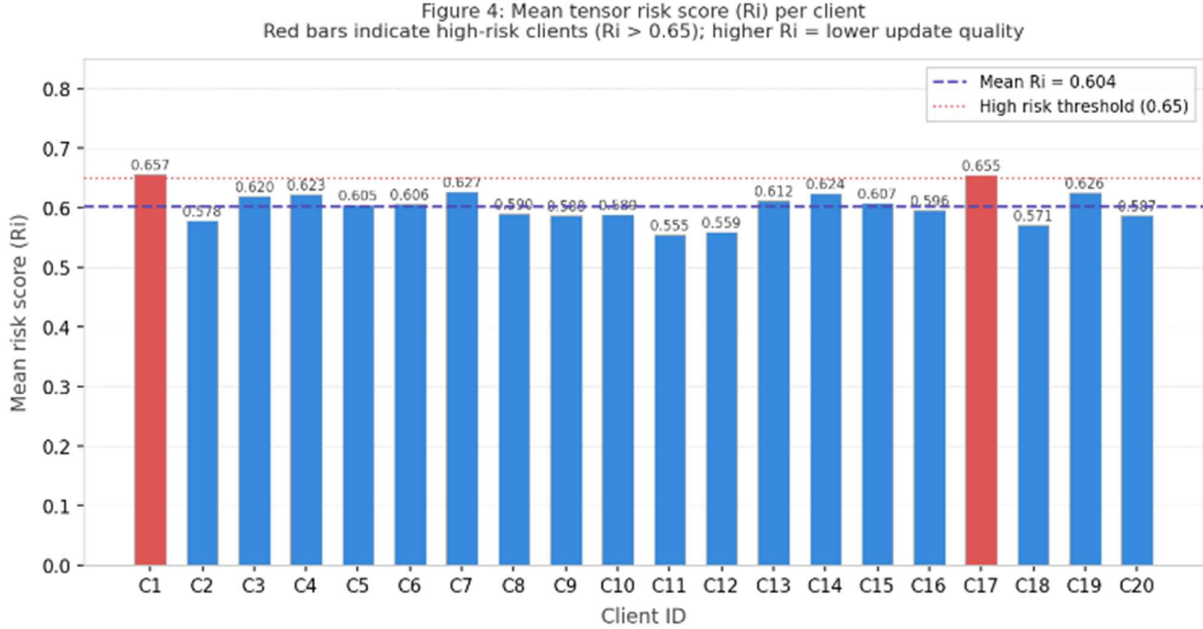


Fig. 4: Mean tensor risk score (R_i) per client. Red bars indicate high-risk clients ($R_i > 0.65$) whose updates exhibit high magnitude deviation or low directional similarity with the global average.

The average score is measured as 0.65. Most clients clustered around this value. From this graph we can clearly see that the clients C1 and C17 have a higher risk tensor. Sandbox testing discussed in Section III-D can be used for clarifying the quality of the tensor.

D. Temporal Score propagation

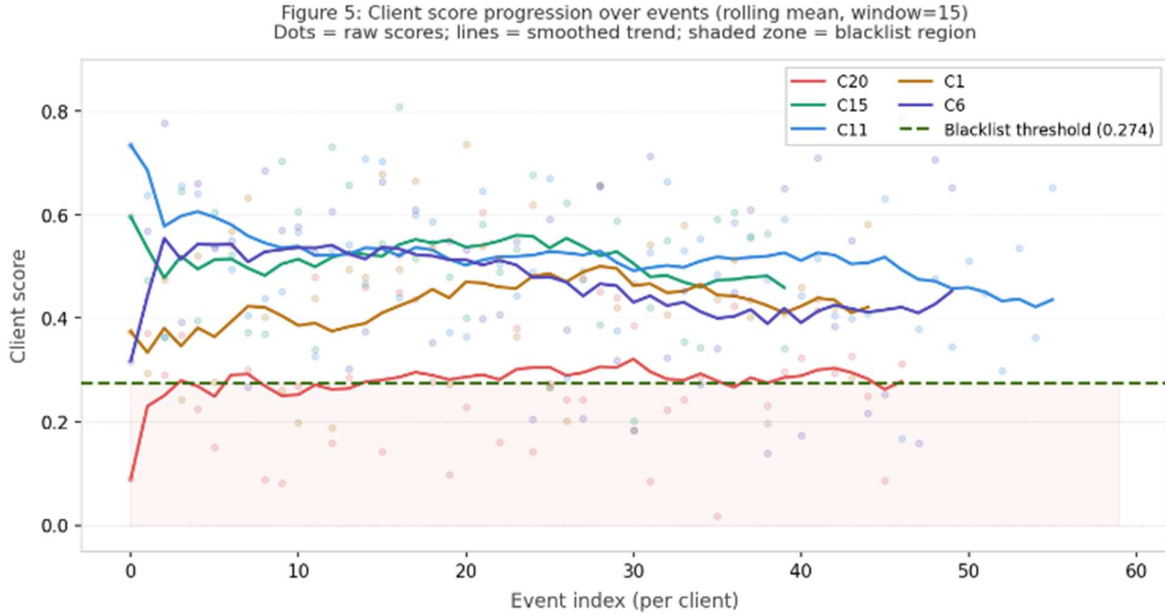


Fig. 5: Score progression of selected clients over simulation events. Lines represent rolling mean (window = 15); dots represent raw scores. The dashed green line marks the blacklisting threshold (0.274); the shaded region indicates the blacklist zone.

The fig. 5 shows that reliable clients maintain stable score above the threshold value. Client C20 demonstrates persistently low scores across events. This confirms long-term unreliable behaviour rather than temporary fluctuation. The time-aware formula introduced in Section III-H captures all the fluctuation in the server.

E. Trust-Aware Aggregation

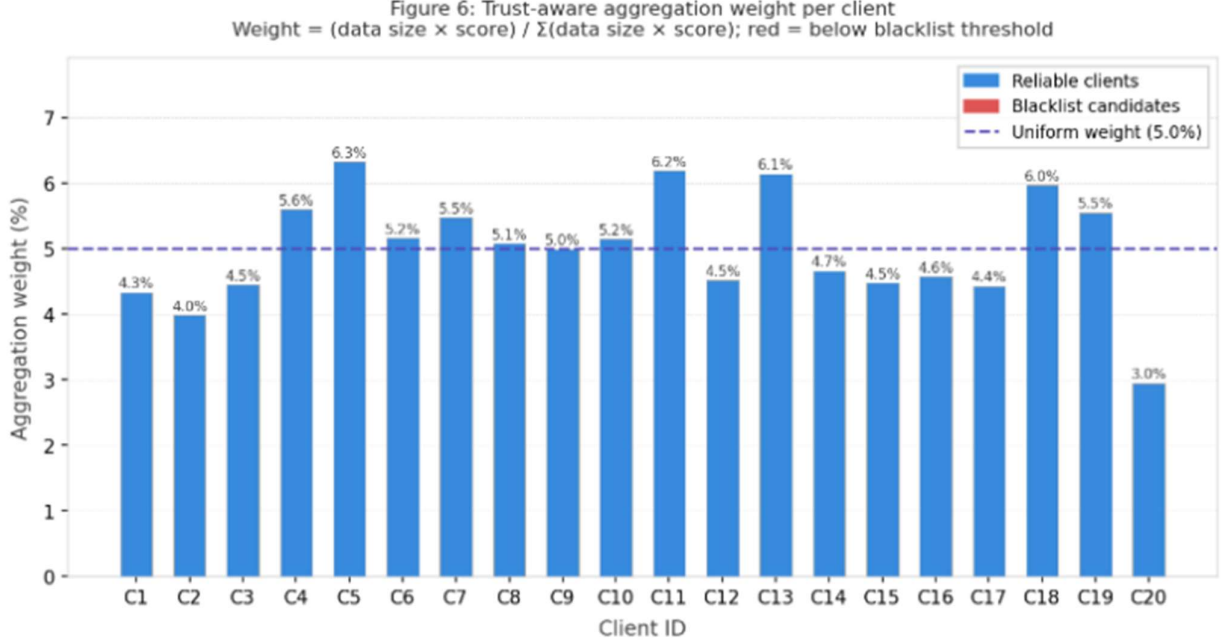


Fig. 6: Trust-aware aggregation weight per client computed as the normalised product of data size and composite score. Red bars indicate clients whose low scores reduce their influence in global model aggregation

The experiment confirms that reliable clients receive higher weight and low-scoring clients receive reduced contribution. Client C20 has the lowest aggregation weight of 3%. The aggregation method proposed in Section III-H effectively integrates data points, client reliability and trust-awareness score through this experiment.

F. Client Score Evaluation

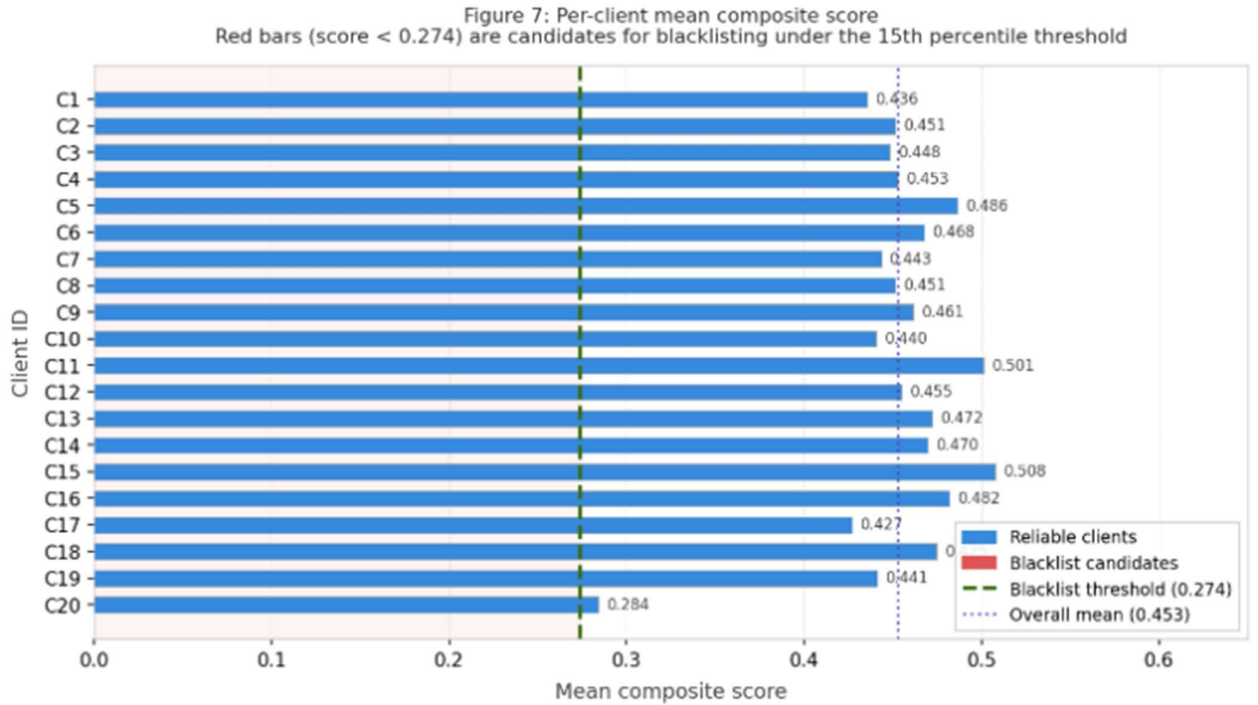


Fig. 7: Mean composite client score across all simulation events.

Majority participants maintain a score between 0.40 to 0.52 range. This confirms that most clients' contribution is stable. Client C20's mean score is 0.284, which is slightly above the threshold value 0.274. That means the scoring mechanism consistently identifies low quality participants.

G. Overall System Behaviour

The experiment demonstrates that the proposed framework efficiently distinguish safe, moderate and unreliable or malicious clients. Handle asynchronous and repeated client updates by preventing malicious sequence. The threshold that found from the experiment can be used to filter out clients without continuous ranking. This act helps to reduce the computational cost of the overall system.

H. Screenshots of Server and Client UI



fig. 8: Server receiving heartbeat signals from the clients

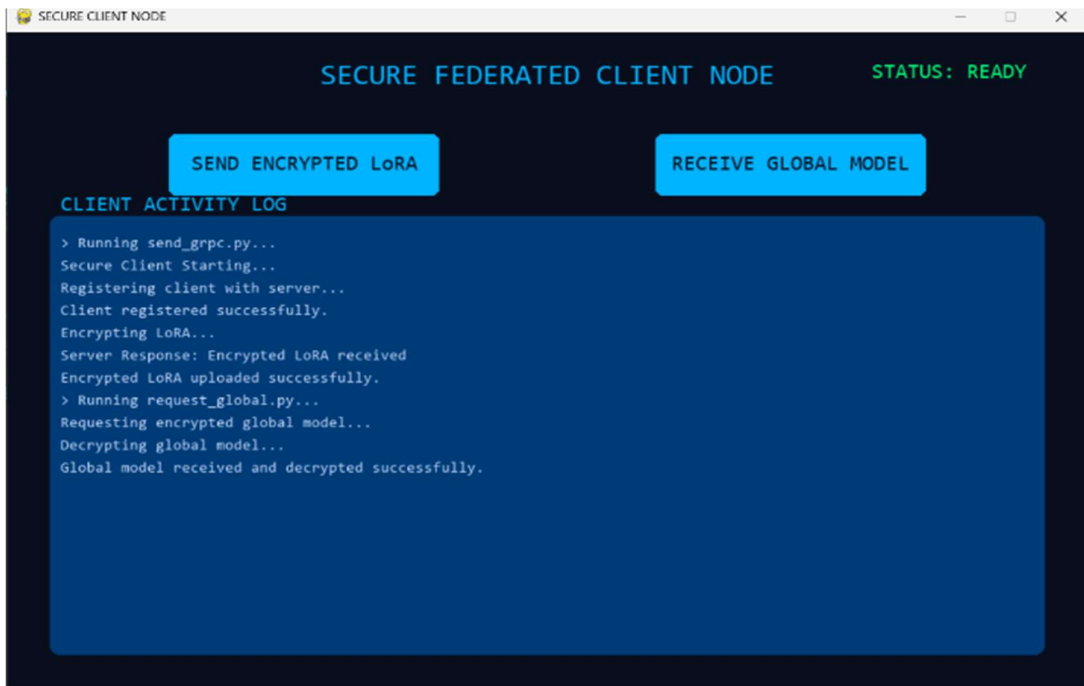


Fig. 9: client sending Local LoRA and requesting Global LoRA

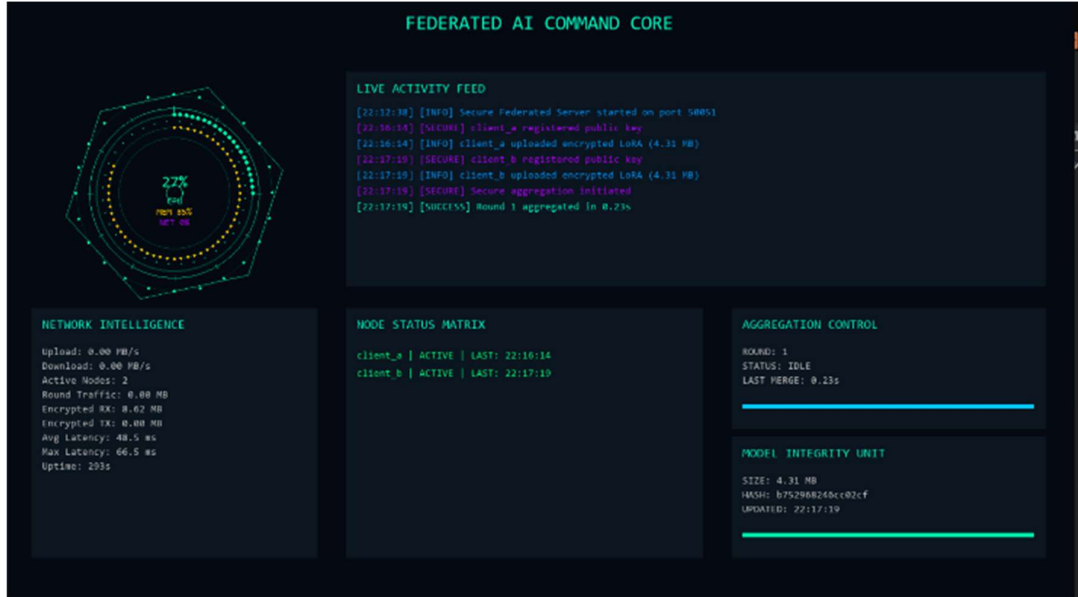


Fig. 10: Server aggregation for clients' updates

Fig. 10 demonstrates the server-side aggregation process, where the server collects and aggregates LoRA updates from different clients. The screenshots confirm that the proposed system operates as intended. Fig. 8 shows the real-time heartbeat monitoring, validating secure and reliable client server communication functionality.

Together, these results validate the proposed framework as a scalable, trust-aware, and privacy-preserving solution for federated medical AI.

VI. LIMITATIONS

The proposed system ensures privacy preserving distributed learning. But several limitations may arise while implementing into real systems. One major limitation is dependency on active client participation. The performance heavily relies on contributions made by multiple clients. A reduced number of participating clients may lead to insufficient model generalization and decrease the accuracy.

Even when the system is using gRPC, the communication overhead is associated with transmitting model updates between clients and server. Network instability, latency or bandwidth constraints can negatively impact the speed and reliability of the aggregation process. Integration of blockchain introduces computational and infrastructural complexity. Multiple validation logs and storing immutable can increase operational cost of the system.

Differential privacy enhances data protection but increase in noise can seriously affect the model performance. Balancing privacy and performance remains a challenge in the system.

VII. FUTURE WORKS

The system can enhance performance and reduce the computational overhead by several methods. One potential improvement lies in blockchain mechanism. Optimizing the blockchain implementation helps to reduce performance issue in validation mechanism. By introducing lightweight consensus algorithms or adaptive validation strategies, the time required for server agreement can be significantly reduced.

Another promising direction is the adoption of advanced aggregation techniques, such as weighted averaging, attention-based aggregation, or trust-aware mechanisms. The system can also be extended to real-world healthcare environments, such as hospitals and diagnostic centers. Deploying the framework with real patient datasets would enable validation of its effectiveness under practical conditions and improve its robustness.

Moreover, the Differential Privacy Retrieval-Augmented Generation (DP-RAG) component can be enhanced by implementing adaptive noise control mechanisms. This would allow a better balance between data privacy and model accuracy, leading to improved performance in sensitive applications.

VIII. REAL-WORLD APPLICATIONS

Researchers can utilize globally trained models derived from diverse datasets to identify disease patterns, improve diagnostics, and develop new treatment strategies. This collaborative approach accelerates innovation while respecting data privacy. The proposed system has wide applicability across various real-world domains, particularly in healthcare. One of the most significant applications is secure hospital data sharing. Multiple hospitals can collaboratively train machine learning models without exposing raw patient data, thereby maintaining confidentiality while benefiting from shared knowledge.

The system is also highly suitable for remote healthcare and telemedicine. Edge devices such as smartphones, wearable sensors, and IoT-enabled medical devices can participate in model training. This enables improved healthcare services in rural and underserved regions. The framework supports personalized medicine by allowing models to adapt to individual patient data. This leads to more accurate predictions, customized treatments, and better patient outcomes.

Beyond healthcare, similar architectures can be applied in finance, smart cities, and cybersecurity, where privacy-preserving collaborative learning is essential.

IX. SECURITY CONSIDERATIONS

The use of blockchain technology enhances transparency and trust by maintaining an immutable record of all transactions and model updates. This ensures accountability and prevents tampering with the aggregation process. Security is a critical component of the proposed system, especially given the sensitivity of medical data. The system employs hybrid encryption techniques to ensure that data remains protected during transmission between clients and servers. This prevents unauthorized access and data leakage.

The tensor tagging mechanism plays a vital role in identifying potentially malicious or low-quality updates from clients. By assigning scores based on behavior and update quality, the system can filter out unreliable contributions, thereby protecting the integrity of the global model. Despite these measures, the system is not immune to emerging threats such as adversarial attacks, model poisoning, or inference attacks. Therefore, continuous monitoring, regular security audits, and adaptive defense mechanisms are necessary to maintain system robustness.

X. SYSTEM ADVANTAGES

The scoring and reputation mechanism helps to identify and filter unreliable or malicious clients. By prioritizing high-quality updates, the system improves the overall performance and stability of the global model. The proposed system provides several significant advantages that make it highly suitable for privacy-sensitive applications. One of the primary benefits is its ability to preserve data privacy by ensuring that raw data never leaves the client device. This is particularly important in healthcare, where data confidentiality is crucial.

Blockchain integration adds layers of transparency and trust by enabling secure and verifiable tracking. This ensures that all participants can rely on the integrity of the system. Furthermore, the combination of federated learning, differential privacy, and secure aggregation creates a robust and efficient framework for distributed intelligence. This makes the system highly adaptable to various domains requiring secure collaborative learning.

XI. CONCLUSION

This work presents a privacy preserving federated learning medical AI in distributed system that addresses the key challenges in data privacy, communication efficiency and reliability. This helps to enable decentralized training of global model using selective LoRA without compromising security and

privacy. This mechanism allows resource-constrained clients to participate in the system. A client scoring mechanism is introduced by combining consistency, tensor-based evaluation and server validation. Blockchain integration helps to ensure servers trust and secure validation. In addition, through differentially private RAG, the system solves the issue of hallucination. FHIR helps with the interoperable handling of medical data. Secure communication is initiated using gRPC and hybrid encryption methodology. The system also suggests a distribution-based client filtering for reducing computational overhead. Future work can focus on optimization of validation and aggregation strategies in large-scale real-world implementation.

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APPENDIX

Event	Client	Server	Heartbeat	m	Hi	Ci_h	Si_t	Ci	Mi		
1	C4	S1	[22, 22, 19, 18, 32]	5	4		1	0.81	0.81	0.4	0.57
2	C3	S1	[31, 34, 15, 32]	4	1	0.333	0.475	0.158	0.52		0.72
3	C8	S2	[15, 20, 28, 25, 23]	5	4		1	0.79	0.79	0.47	0.92
4	C13	S1	[26, 34, 23, 16, 29]	5	2	0.667	0.64	0.427	0.89		0.93
5	C10	S3	[33, 21, 17, 16, 22]	5	4		1	0.77	0.77	1.15	0.94
6	C13	S2	[35, 26, 20, 26, 26, 21]	6	2	0.667	0.717	0.478	1.04		0.76
7	C6	S3	[20, 29, 27, 23]	4	2	0.667	0.762	0.508	1.39		0.72
8	C11	S1	[16, 25, 27, 23]	4	3		1	0.762	0.762	0.37	0.89
9	C16	S2	[19, 23, 19, 22, 32, 32]	6	4		1	0.742	0.742	0.59	0.68
10	C5	S3	[17, 16, 18, 19, 35, 20]	6	5		1	0.792	0.792	1.17	0.57
11	C13	S2	[31, 23, 32, 15, 18, 32]	6	3		1	0.625	0.625	1.13	0.8
12	C14	S1	[15, 23, 31, 20, 31, 18]	6	4		1	0.733	0.733	1.26	0.49
13	C6	S3	[34, 25, 30]	3	1	0.333	0.517	0.172	0.32		0.9
14	C8	S1	[33, 17, 17, 30]	4	2	0.667	0.637	0.425	1.2		0.94
15	C16	S3	[23, 31, 34, 28]	4	1	0.333	0.55	0.183	1.36		0.9
16	C13	S3	[29, 31, 29, 18, 22]	5	2	0.667	0.67	0.447	0.55		0.52
17	C1	S1	[22, 17, 16]	3	3		1	0.85	0.85	1.25	0.35
18	C9	S3	[21, 32, 19, 33, 33, 30]	6	2	0.667	0.583	0.389	0.57		0.61
19	C4	S3	[26, 28, 28, 29, 16, 35]	6	1	0.333	0.583	0.194	1.38		0.36
20	C11	S1	[21, 21, 32, 29]	4	2	0.667	0.713	0.475	0.45		0.42
21	C3	S2	[16, 35, 32]	3	1	0.333	0.483	0.161	1.22		0.93
22	C6	S2	[30, 21, 27, 16, 20, 27]	6	3		1	0.758	0.758	0.3	0.55
23	C10	S2	[19, 21, 24, 21, 16, 33]	6	5		1	0.8	0.8	1.11	0.34

24	C19	S2	[16, 31, 17, 20]	4	3	1	0.775	0.775	0.38	0.34
25	C19	S1	[34, 17, 28]	3	1	0.333	0.583	0.194	1.02	0.67
26	C7	S3	[22, 23, 27, 19, 35]	5	3	1	0.72	0.72	0.63	0.51
27	C20	S3	[]	0	0	0	0	0	1.4	0.35
28	C5	S2	[22, 26, 24]	3	2	0.667	0.8	0.533	0.47	0.84
29	C20	S3	[]	0	0	0	0	0	0.88	0.73
30	C4	S1	[18, 18, 32, 19, 23]	5	4	1	0.8	0.8	0.61	0.44
31	C7	S3	[31, 30, 23, 16, 17]	5	3	1	0.69	0.69	1	0.84
32	C5	S3	[20, 29, 32, 28, 32]	5	1	0.333	0.59	0.197	0.31	0.35
33	C18	S1	[33, 32, 19, 28, 19]	5	2	0.667	0.65	0.433	0.35	0.54
34	C7	S3	[18, 26, 32, 28]	4	1	0.333	0.65	0.217	1.37	0.79
35	C6	S2	[20, 25, 28]	3	2	0.667	0.783	0.522	1.18	0.86
36	C9	S1	[27, 16, 30]	3	1	0.333	0.65	0.217	0.54	0.83
37	C8	S1	[21, 27, 25]	3	2	0.667	0.783	0.522	0.61	0.35
38	C17	S2	[15, 18, 23, 20, 33]	5	4	1	0.77	0.77	1.36	0.47
39	C12	S3	[28, 34, 31, 18, 27]	5	1	0.333	0.58	0.193	1.29	0.42
40	C1	S3	[26, 28, 17, 25]	4	2	0.667	0.725	0.483	0.99	0.73
41	C10	S3	[28, 25, 27, 24, 32]	5	2	0.667	0.64	0.427	0.44	0.57
42	C6	S3	[27, 32, 15, 24, 24]	5	3	1	0.68	0.68	0.53	0.81
43	C15	S2	[21, 31, 30, 20, 17, 24]	6	4	1	0.758	0.758	0.87	0.71
44	C8	S3	[22, 21, 19, 15, 16]	5	5	1	0.87	0.87	0.57	0.61
45	C19	S1	[30, 27, 22, 19, 35, 15]	6	3	1	0.667	0.667	1.28	0.86
46	C8	S1	[16, 32, 22, 18, 29, 19]	6	4	1	0.75	0.75	1.18	0.73
47	C15	S3	[32, 29, 20, 30, 29, 23]	6	2	0.667	0.642	0.428	1.13	0.85
48	C17	S2	[23, 29, 17, 24]	4	3	1	0.762	0.762	0.56	0.52
49	C8	S2	[21, 17, 28, 28]	4	2	0.667	0.75	0.5	0.66	0.6
50	C14	S2	[33, 27, 30]	3	0	0	0.5	0	0.31	0.53
51	C14	S3	[30, 22, 23, 28]	4	2	0.667	0.713	0.475	0.83	0.55
52	C15	S1	[27, 33, 33]	3	0	0	0.45	0	1.03	0.35
53	C15	S1	[23, 27, 25]	3	2	0.667	0.75	0.5	0.53	0.51
54	C14	S2	[30, 15, 32]	3	1	0.333	0.55	0.183	0.36	0.92
55	C2	S1	[21, 15, 34, 19]	4	3	1	0.738	0.738	0.56	0.61
56	C19	S1	[23, 26, 20, 34, 34, 18]	6	3	1	0.675	0.675	1.16	0.41
57	C1	S2	[27, 21, 17, 33, 35, 22]	6	3	1	0.658	0.658	0.41	0.8
58	C19	S1	[32, 28, 26, 17, 31]	5	1	0.333	0.6	0.2	1.01	0.31
59	C4	S2	[35, 29, 19, 28, 20]	5	2	0.667	0.67	0.447	1.11	0.93
60	C20	S3	[]	0	0	0	0	0	1.15	0.6
61	C8	S1	[29, 22, 29, 33, 34]	5	1	0.333	0.53	0.177	1.04	0.52
62	C6	S2	[26, 23, 25, 23]	4	3	1	0.787	0.787	1.27	0.76
63	C18	S1	[17, 22, 28, 30]	4	2	0.667	0.713	0.475	0.91	0.46
64	C16	S2	[17, 24, 22]	3	3	1	0.85	0.85	0.74	0.46
65	C17	S2	[32, 25, 26, 29, 23, 24]	6	3	1	0.675	0.675	0.58	0.38
66	C18	S3	[21, 21, 30, 23]	4	3	1	0.812	0.812	1.1	0.93

67	C7	S2	[26, 20, 24, 15]	4	3	1	0.812	0.812	1.08	0.38
68	C18	S2	[35, 30, 18, 15]	4	2	0.667	0.6	0.4	0.93	0.61
69	C2	S2	[18, 17, 27, 30, 17, 33]	6	3	1	0.683	0.683	0.99	0.33
70	C19	S2	[22, 18, 32]	3	2	0.667	0.733	0.489	1.14	0.69
71	C13	S2	[24, 33, 28, 24, 33, 34]	6	2	0.667	0.533	0.356	0.37	0.92
72	C7	S3	[23, 17, 20, 22]	4	4	1	0.9	0.9	0.49	0.35
73	C20	S2	[]	0	0	0	0	0	0.62	0.45
74	C15	S1	[23, 35, 33, 21]	4	2	0.667	0.6	0.4	0.77	0.65
75	C5	S1	[20, 24, 34]	3	2	0.667	0.7	0.467	1.12	0.67
76	C15	S1	[24, 27, 23, 31, 32, 30]	6	2	0.667	0.608	0.406	0.78	0.69
77	C20	S2	[]	0	0	0	0	0	0.33	0.45
78	C9	S3	[20, 30, 31]	3	1	0.333	0.65	0.217	1.02	0.9
79	C14	S3	[17, 30, 26, 28, 25, 25]	6	3	1	0.692	0.692	1.04	0.86
80	C16	S2	[32, 16, 29, 17, 25, 23]	6	4	1	0.7	0.7	0.66	0.93
81	C1	S3	[28, 16, 21, 31, 26, 34]	6	2	0.667	0.633	0.422	1.13	0.71
82	C7	S2	[24, 29, 30, 18]	4	2	0.667	0.688	0.458	0.33	0.71
83	C10	S3	[32, 28, 17]	3	1	0.333	0.617	0.206	0.55	0.85
84	C4	S3	[30, 24, 31, 23]	4	2	0.667	0.65	0.433	0.76	0.61
85	C8	S2	[27, 21, 34, 31]	4	1	0.333	0.588	0.196	1.12	0.39
86	C14	S2	[15, 24, 24, 33, 33]	5	3	1	0.61	0.61	1.4	0.62
87	C16	S2	[32, 32, 27, 29, 25]	5	1	0.333	0.55	0.183	1.26	0.94
88	C8	S2	[25, 30, 27]	3	1	0.333	0.633	0.211	0.72	0.73
89	C2	S1	[18, 29, 18, 31, 29]	5	2	0.667	0.67	0.447	0.32	0.39
90	C3	S2	[25, 34, 27, 35, 17]	5	2	0.667	0.56	0.373	1.24	0.85
91	C11	S3	[32, 16, 34, 17, 22, 35]	6	3	1	0.583	0.583	1.05	0.49
92	C14	S1	[29, 20, 24]	3	2	0.667	0.783	0.522	1.29	0.33
93	C12	S2	[22, 31, 28, 33]	4	1	0.333	0.575	0.192	1.05	0.42
94	C13	S3	[30, 33, 19, 22]	4	2	0.667	0.675	0.45	0.81	0.47
95	C15	S2	[17, 29, 26, 33]	4	1	0.333	0.613	0.204	0.63	0.93
96	C10	S1	[30, 18, 22, 27, 33, 26]	6	2	0.667	0.667	0.444	0.93	0.95
97	C13	S2	[33, 16, 34]	3	1	0.333	0.483	0.161	1.12	0.84
98	C8	S3	[22, 35, 21, 34, 23]	5	3	1	0.65	0.65	1.05	0.77
99	C2	S2	[16, 33, 26, 19, 17, 24]	6	4	1	0.742	0.742	0.66	0.57